

1. (15 points) An insulated cylinder contains 5.0 liters of air at room temperature (300 K) and atmospheric pressure. (10^5 N/m^2). At one end of the cylinder is a massless piston, whose surface area is 0.01 m^2 . Suppose that you push the piston in very hard, exerting a force of 2500 N. The piston moves 4 mm before it is stopped.

- a. (5 points) Calculate the amount of work you did on the system, and the heat added to the gas. Assuming that all of the energy added goes into the gas (not the piston or cylinder walls), by how much does the internal energy of the gas increase?

$$W = Fd = (2500 \text{ N}) (4 \times 10^{-3} \text{ m}) = 10 \text{ J} \checkmark$$

$$Q = 0: \text{ no heat transfer } \checkmark$$

$$\Delta U = Q + W = 0 + 10 \text{ J} = 10 \text{ J} \checkmark$$

- b. (5 points) Calculate the change in the entropy of the gas once it has again reached equilibrium. (Because the piston displacement is tiny compared with the total volume of gas, you may assume the internal pressure and temperature of the gas are constant in this problem).

$$dU = TdS - PdV + \mu dN$$

$$\text{same amt of gas: } dN = 0$$

$$dS = \frac{dU}{T} + \frac{P}{T} dV \quad \text{sign -}$$

$$dV = V_f - V_i$$

$$\Delta S = \frac{10 \text{ J}}{300 \text{ K}} + \frac{10^5 \text{ N/m}^2}{300 \text{ K}} (4 \times 10^{-5} \text{ m}^3)$$

$$\text{or } (0.01 \text{ m}^2) (4 \times 10^{-3} \text{ m}) = 4 \times 10^{-5} \text{ m}^3$$

$$= \frac{14 \text{ J}}{300 \text{ K}} = 4.67 \times 10^{-2} \text{ J/K}$$

- c. (5 points) Is this process reversible or irreversible? Explain. If it is reversible, how could you make it irreversible? Likewise, if it is irreversible, how could you make it reversible?

It is irreversible: $\Delta S > 0$. More energy was imparted to the gas than necessary to achieve the volume change, and removing the piston will not recover that extra energy.

If it is to be reversible, the piston must be pushed in quasistatically.

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2. (15 points) The multiplicity for a cold, large Einstein solid is:

$$\Omega \approx \left(\frac{eN}{q} \right)^q$$

With the system's total energy defined by:

$$U = \epsilon q$$

$$q = \frac{U}{\epsilon}$$

(a) (5 points) Calculate the temperature of the solid as a function of its total energy.

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$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{N,V}$$

$$S = k \ln \Omega = k \ln \left[\left(\frac{eN}{U/\epsilon} \right)^{U/\epsilon} \right]$$

$$= \frac{kU}{\epsilon} \left[\ln(eN) - \ln\left(\frac{U}{\epsilon}\right) \right]$$

$$\begin{aligned} \frac{\partial S}{\partial U} &= \frac{k}{\epsilon} \ln(eN) - \frac{k}{\epsilon} \ln\left(\frac{U}{\epsilon}\right) - \frac{kU}{\epsilon} \left(\frac{1}{U/\epsilon} \right) \left(\frac{1}{\epsilon} \right) \\ &= \frac{k}{\epsilon} \left[\ln(eN) - \ln\left(\frac{U}{\epsilon}\right) \right] - \frac{k}{\epsilon} \end{aligned}$$

$$T = \frac{1}{\frac{k}{\epsilon} \left[\ln(eN) - \ln\left(\frac{U}{\epsilon}\right) \right] - \frac{k}{\epsilon}} \quad \text{clean up}$$

(b) (5 points) Calculate the energy of the solid as a function of temperature.

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$$\frac{1}{T} = \frac{k}{\epsilon} \left[\ln(eN) - \ln\left(\frac{U}{\epsilon}\right) \right] - \frac{k}{\epsilon}$$

$$\frac{1}{T} + \frac{k}{\epsilon} - \frac{k}{\epsilon} \ln(eN) = -\frac{k}{\epsilon} \ln\left(\frac{U}{\epsilon}\right)$$

$$\ln\left(\frac{U}{\epsilon}\right) = -\frac{\epsilon}{kT} + (-1) + \ln(eN)$$

$$\begin{aligned} \frac{U}{\epsilon} &= \exp\left(-\frac{\epsilon}{kT} - 1 + \ln(eN)\right) \\ &= e^{-(\epsilon/kT + 1)} e^{\ln(eN)} = e^{-(\epsilon/kT + 1)} eN \end{aligned}$$

$$U(T) = eN\epsilon \exp\left(-\frac{\epsilon}{kT} - 1\right) \quad 5$$

- (c) (5 points) Calculate the constant volume heat capacity of the solid.

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$$C_v = \left(\frac{\partial U}{\partial T} \right)_V$$

$$U(T) = (eNe) \exp\left(-\frac{e}{kT} - 1\right)$$

$$\frac{\partial U}{\partial T} = (eNe) \exp\left(-\frac{e}{kT} - 1\right) \left(\frac{e}{kT^2}\right)$$

$$C_v = \frac{e^2 Ne}{kT^2} \exp\left(-\frac{e}{kT} - 1\right)$$

3. (15 points) Calcium carbonate has two common crystalline forms, calcite and aragonite, with the following thermodynamic data:

$\text{cm}^3 \times \frac{\text{m}^3}{1000 \text{ cm}^3}$

Form	$\Delta_f H$ (kJ)	$\Delta_f G$ (kJ)	S (J/K)	V (cm ³)
calcite	-1206.9	-1128.8	92.9	36.93
aragonite	-1207.1	-1127.8	88.7	34.15

All data are for one mole of material at 298 K and 1 bar. S and V can be considered independent of temperature for this problem.

$$T = 298 \text{ K} \quad P = 10^5 \text{ Pa}$$

$$H = U + PV$$

$$U = H - PV$$

$$G = U - TS + PV$$

$$H = G + TS$$

- a. (5 points) Which form is stable at the earth's surface and why?

$$\Delta S_{\text{calcite}} = \frac{1}{T} (-(-1128.8 \times 10^3 \text{ J}) + (-1206.9 \text{ kJ})) = -262 \text{ J/K}$$

$$\Delta S_{\text{aragonite}} = \frac{1}{298 \text{ K}} (-(-1127.8) + (-1207.1)) = -265 \text{ J/K}$$

$$\Delta U_{\text{calcite}} = -1128.8 \times 10^3 \text{ J} + (298)(92.9) - (10^5 \text{ Pa})(36.93 \times 10^{-6} \text{ m}^3) = -1101.1 \text{ kJ}$$

$$\Delta U_{\text{aragonite}} = -1127.8 \times 10^3 \text{ J} + (298)(88.7) - (10^5 \text{ Pa})(34.15 \times 10^{-6} \text{ m}^3) = -1101.3 \text{ kJ}$$

$$G = H - TS$$

$$U = G + TS - PV$$

- b. (5 points) Assume the temperature is kept at room temperature. Calculate the pressure at which the phase not stable in part a should become stable.

$$\Delta U = \Delta G$$

Aragonite:

$$TS = PV$$

$$P = \frac{TS}{V} = \frac{(298)(88.7)}{34.15 \times 10^{-6}} = 7.75 \times 10^8 \text{ Pa}$$

Calcite is stable at the Earth's surface; both entropy and total energy are lower.